

Friday, May 3

This is part of an [ongoing research project](#) in collaboration with researchers in the College of Natural Resources.

Setup

The following packages are being used.

```
library(dplyr)      # data manipulation
library(tidyr)      # data manipulation
library(lubridate)  # working with dates
library(forcats)    # working with factors
library(wherami)    # find current directory
library(ggplot2)    # graphics
library(mgcv)       # GAMs
library(trtools)    # inference tools
library(emmeans)    # inference tools
options(digits = 4) # control number of digits displayed
```

Import and Process Tick Count Data

Here I can use `dirname(thisfile())` to find the directory containing this Rmarkdown file, so I do not have to specify the full path to the data file. Note that `thisfile()` is from the **wherami** package. I have the data stored in a sub-directory ("tickdata") of the directory containing this Rmarkdown file.

```
ticks <- read.csv(paste(dirname(thisfile()),
  "/tickdata/tick_data_Robenstein.csv", sep = ""))
names(ticks) <- c("moose", "mortality", "ticks", "size", "date", "gmu", "sex", "note")
head(ticks)
```

	moose	mortality	ticks	size	date	gmu	sex	note
1	21005370	H	0	100	9/16/2020	1	MALE	1
2	21005396	H	21	100	10/14/2020	1	MALE	1
3	21005452	H	0	100	10/5/2020	1	MALE	1
4	21005506	H	9	100	11/11/2020	1	MALE	1
5	21005526	H	34	100	11/22/2020	1	MALE	1
6	21005538	H	1	100	11/5/2020	1	MALE	0

Here I am going to process the data to get it ready for plotting and modeling.

```
ticks <- ticks %>%
  mutate(note = factor(note, levels = 0:3,
    labels = c("exclude", "include", "deterioration", "nodate"))) %>%
  mutate(date = mdy(date)) %>%
  mutate(month = month(date, label = TRUE), year = year(date), day = yday(date)) %>%
  filter(!is.na(date), month %in% c("Aug", "Sep", "Oct", "Nov")) %>%
  mutate(year = factor(year)) %>% mutate(sex = tolower(sex)) %>%
  mutate(day = ifelse(year == "2020", day - 1, day))
head(ticks)
```

	moose	mortality	ticks	size	date	gmu	sex	note	month	year	day	
1	21005370		H	0	100	2020-09-16	1	male	include	Sep	2020	259
2	21005396		H	21	100	2020-10-14	1	male	include	Oct	2020	287
3	21005452		H	0	100	2020-10-05	1	male	include	Oct	2020	278
4	21005506		H	9	100	2020-11-11	1	male	include	Nov	2020	315
5	21005526		H	34	100	2020-11-22	1	male	include	Nov	2020	326
6	21005538		H	1	100	2020-11-05	1	male	exclude	Nov	2020	309

Import and Process Game Management Unit Data

Here I am going to use `rename` to rename the imported variables.

```
gmu <- read.csv(paste(dirname(thisfile()), "/tickdata/tick_study_areas.csv", sep = "")) %>%
  rename(gmu = GMU, area = study.Area, samples = hh_samples)
head(gmu, 10)
```

	gmu	area	samples
1	1	North	27
2	2	North	36
3	3	North	19
4	4	North	24
5	4A	North	6
6	5 North Central		18
7	6 North Central		22
8	7 North Central		7
9	8 North Central		9
10	8A North Central		8

Merging Data Frames and Filtering

```
ticks <- left_join(ticks, gmu) %>%
  mutate(area = factor(area)) %>%
  mutate(area = fct_relevel(area, c("North", "North Central",
    "Central", "Southeast", "Island Park")))
head(ticks)
```

	moose	mortality	ticks	size	date	gmu	sex	note	month	year	day	area	samples
1	21005370		H	0	100	2020-09-16	1	male	include	Sep	2020	North	27
2	21005396		H	21	100	2020-10-14	1	male	include	Oct	2020	North	27
3	21005452		H	0	100	2020-10-05	1	male	include	Oct	2020	North	27
4	21005506		H	9	100	2020-11-11	1	male	include	Nov	2020	North	27
5	21005526		H	34	100	2020-11-22	1	male	include	Nov	2020	North	27
6	21005538		H	1	100	2020-11-05	1	male	exclude	Nov	2020	North	27

Some variables we do not need. Also we are going to discard some questionable observations and focus only on the males.

```
ticks <- ticks %>% select(-mortality, -samples) %>%
  filter(note == "include") %>% filter(sex == "male")
head(ticks)
```

	moose	ticks	size	date	gmu	sex	note	month	year	day	area
1	21005370	0	100	2020-09-16	1	male	include	Sep	2020	259	North
2	21005396	21	100	2020-10-14	1	male	include	Oct	2020	287	North
3	21005452	0	100	2020-10-05	1	male	include	Oct	2020	278	North
4	21005506	9	100	2020-11-11	1	male	include	Nov	2020	315	North

5	21005526	34	100	2020-11-22	1 male include	Nov 2020	326	North
6	21005546	1	100	2020-11-22	1 male include	Nov 2020	326	North

Raw Data Visualization

```
d09 <- yday(mdy("09/1/2021"))
d10 <- yday(mdy("10/1/2021"))
d11 <- yday(mdy("11/1/2021"))
d12 <- yday(mdy("12/1/2021"))

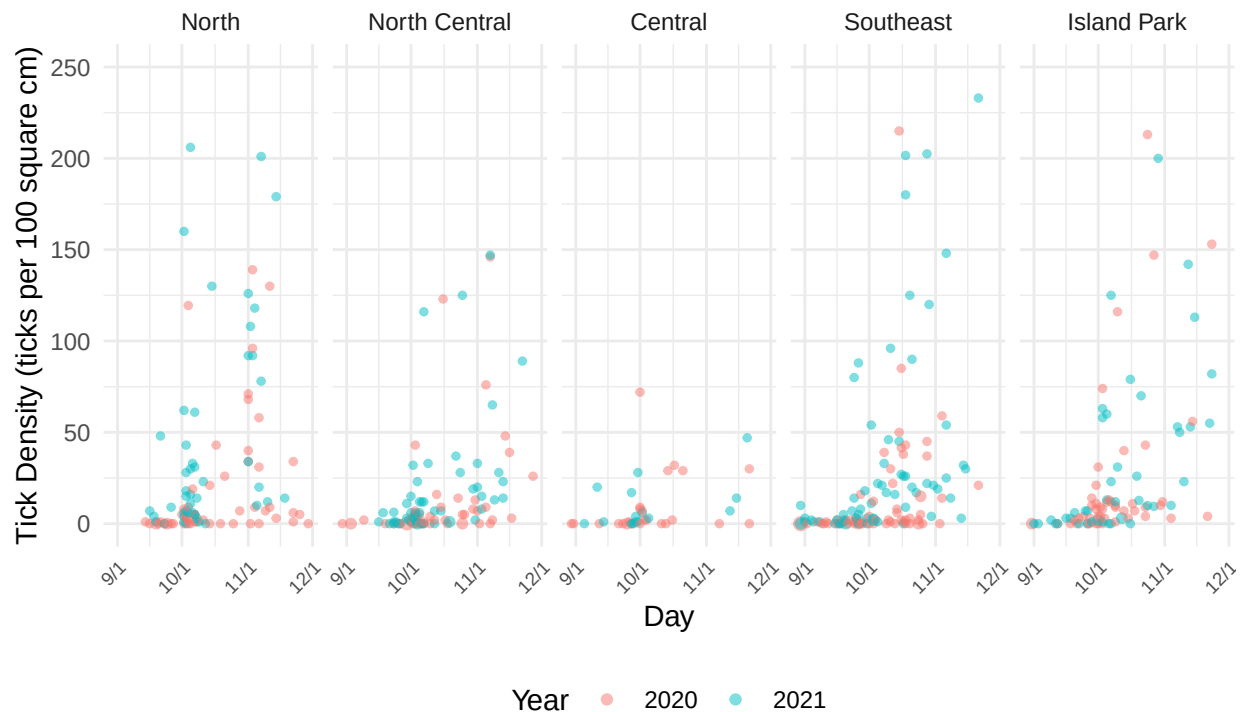
pv <- ggplot(ticks, aes(x = day, y = ticks/size*100, color = year)) +
  geom_count(alpha = 0.5) + scale_size_area(max_size = 2) +
  theme_minimal() + facet_grid(area ~ .) +
  labs(x = "Day", y = "Tick Density (ticks per 100 square cm)", color = "Year") +
  guides(size = "none") +
  scale_x_continuous(breaks = c(d09, d10, d11, d12),
    labels = c("9/1", "10/1", "11/1", "12/1")) + ylim(c(0,250)) +
  theme(legend.position = "bottom")

plot(pv)
```



```
ph <- ggplot(ticks, aes(x = day, y = ticks/size*100, color = year)) +
  geom_count(alpha = 0.5) + scale_size_area(max_size = 2) + theme_minimal() +
  facet_wrap(~ area, ncol = 5) +
  labs(x = "Day", y = "Tick Density (ticks per 100 square cm)", color = "Year") +
  guides(size = "none") +
  scale_x_continuous(breaks = c(d09, d10, d11, d12),
    labels = c("9/1", "10/1", "11/1", "12/1")) + ylim(c(0,250)) +
  theme(legend.position = "bottom",
    axis.text.x = element_text(angle = 45, size = 7, hjust = 1))

plot(ph)
```



Modeling

I used a generalized additive model with a log link function estimated using (penalized) quasi-likelihood to deal with considerable over-dispersion.

```
m <- gam(ticks ~ offset(log(size)) + s(day) + year + area,
  family = quasipoisson(link = log), data = ticks)
summary(m)
```

Family: quasipoisson
Link function: log

Formula:
ticks ~ offset(log(size)) + s(day) + year + area

Parametric coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-1.9941	0.1849	-10.78	< 2e-16 ***

year2021	0.5734	0.1493	3.84	0.00014 ***
areaNorth Central	-0.6751	0.2304	-2.93	0.00355 **
areaCentral	-0.7342	0.4097	-1.79	0.07376 .
areaSoutheast	-0.0457	0.1923	-0.24	0.81228
areaIsland Park	0.0282	0.2019	0.14	0.88909

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Approximate significance of smooth terms:

	edf	Ref.df	F	p-value
s(day)	2.99	3.78	20.2	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

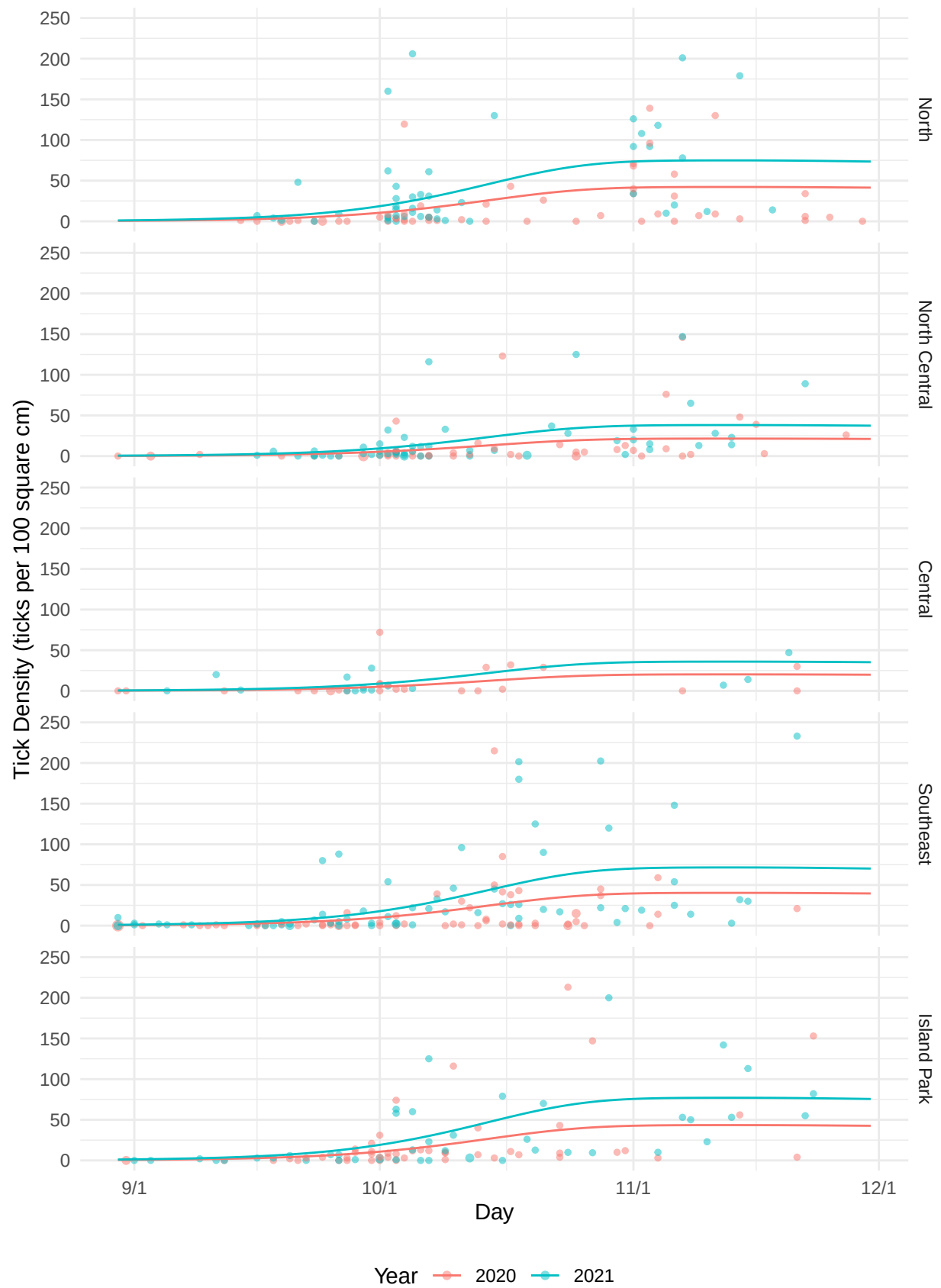
R-sq.(adj) = 0.196 Deviance explained = 33.9%

GCV = 34.395 Scale est. = 54.637 n = 471

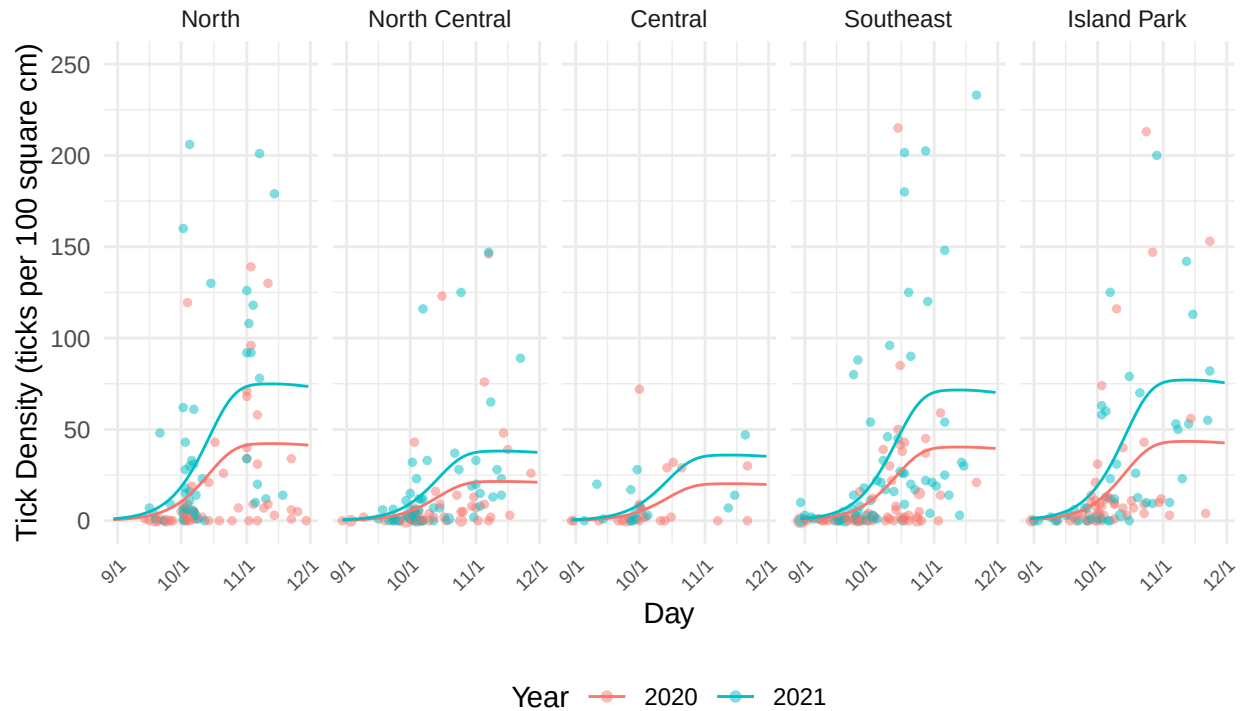
Here we can visualize this model.

```
d <- expand.grid(year = c("2020", "2021"), day = 242:334,
  area = unique(ticks$area), size = 100)
d$yhat <- predict(m, newdata = d, type = "response")

pv <- pv + geom_line(aes(y = yhat), data = d)
plot(pv)
```



```
ph <- ph + geom_line(aes(y = yhat), data = d)
plot(ph)
```



Model-Based Inferences

How much higher is the expected tick density in 2021 than in 2020?

```
emmeans(m, ~year | area, at = list(day = d10),
  type = "response", offset = log(100), data = ticks)
```

area = North:

year	rate	SE	df	lower.CL	upper.CL
2020	10.49	2.14	462	7.02	15.68
2021	18.61	3.61	462	12.71	27.25

area = North Central:

year	rate	SE	df	lower.CL	upper.CL
2020	5.34	1.31	462	3.30	8.64
2021	9.48	2.16	462	6.05	14.83

area = Central:

year	rate	SE	df	lower.CL	upper.CL
2020	5.03	2.04	462	2.27	11.16
2021	8.93	3.61	462	4.03	19.78

area = Southeast:

year	rate	SE	df	lower.CL	upper.CL
2020	10.02	2.12	462	6.62	15.17
2021	17.78	3.39	462	12.22	25.86

area = Island Park:

year	rate	SE	df	lower.CL	upper.CL
2020	10.79	2.32	462	7.07	16.47
2021	19.14	3.77	462	13.00	28.19

Confidence level used: 0.95

Intervals are back-transformed from the log scale

```
pairs(emmeans(m, ~year | area, at = list(day = d10),
  type = "response", offset = log(100), data = ticks), reverse = TRUE, infer = TRUE)
```

area = North:

contrast	ratio	SE	df	lower.CL	upper.CL	null	t.ratio	p.value
year2021 / year2020	1.77	0.265	462	1.32	2.38	1	3.841	0.0001

area = North Central:

contrast	ratio	SE	df	lower.CL	upper.CL	null	t.ratio	p.value
year2021 / year2020	1.77	0.265	462	1.32	2.38	1	3.841	0.0001

area = Central:

contrast	ratio	SE	df	lower.CL	upper.CL	null	t.ratio	p.value
year2021 / year2020	1.77	0.265	462	1.32	2.38	1	3.841	0.0001

area = Southeast:

contrast	ratio	SE	df	lower.CL	upper.CL	null	t.ratio	p.value
year2021 / year2020	1.77	0.265	462	1.32	2.38	1	3.841	0.0001

area = Island Park:

contrast	ratio	SE	df	lower.CL	upper.CL	null	t.ratio	p.value
year2021 / year2020	1.77	0.265	462	1.32	2.38	1	3.841	0.0001

Confidence level used: 0.95

Intervals are back-transformed from the log scale

Tests are performed on the log scale

Note: Here `emmeans` needs a bit more information that is not contained in the model object, so we pass it the data with `data = ticks`.

How about the *difference* in the expected density for each area and the first of October? This is a *discrete marginal effect*, and both area and day matter.

```
margeff(m,
  a = list(year = "2021", area = unique(ticks$area), day = d10, size = 100),
  b = list(year = "2020", area = unique(ticks$area), day = d10, size = 100),
  cnames = unique(ticks$area), df = m$df.residual)
```

	estimate	se	lower	upper	tvalue	df	pvalue
North	8.122	2.546	3.1197	13.125	3.191	462	0.001517
North Central	4.135	1.356	1.4702	6.800	3.049	462	0.002426
Central	3.898	1.865	0.2326	7.563	2.090	462	0.037182
Southeast	7.760	2.351	3.1405	12.379	3.301	462	0.001037
Island Park	8.354	2.574	3.2971	13.412	3.246	462	0.001254

Interestingly this can also be done using the `emmeans` package through use of the `regrid` function.

```
tmp <- emmeans(m, ~year | area, at = list(day = d10),
  type = "response", offset = log(100), data = ticks)
pairs(regrid(tmp, type = "response"), reverse = TRUE, infer = TRUE)
```

```
area = North:
  contrast      estimate    SE  df lower.CL upper.CL t.ratio p.value
year2021 - year2020      8.12 2.55 462    3.120    13.12   3.191 0.0015
```

```
area = North Central:
  contrast      estimate    SE  df lower.CL upper.CL t.ratio p.value
year2021 - year2020      4.13 1.36 462    1.470     6.80   3.049 0.0024
```

```
area = Central:
  contrast      estimate    SE  df lower.CL upper.CL t.ratio p.value
year2021 - year2020      3.90 1.87 462    0.233     7.56   2.090 0.0372
```

```
area = Southeast:
  contrast      estimate    SE  df lower.CL upper.CL t.ratio p.value
year2021 - year2020      7.76 2.35 462    3.141    12.38   3.301 0.0010
```

```
area = Island Park:
  contrast      estimate    SE  df lower.CL upper.CL t.ratio p.value
year2021 - year2020      8.35 2.57 462    3.297    13.41   3.246 0.0013
```

Confidence level used: 0.95

How about inferences for the *average* difference across areas?

```
emmeans(regrid(pairs(regrid(tmp, type = "response"), reverse = TRUE)), ~1)
```

```
1      estimate    SE  df lower.CL upper.CL
overall    6.45 1.85 462    2.82    10.1
```

Results are averaged over the levels of: area

Confidence level used: 0.95

Tricky!

How much does the expected tick density increase between, say, the first day of October and November?

```
emmeans(m, ~ day | year * area, at = list(day = c(d11,d10)),
  data = ticks, type = "response")
```

```
year = 2020, area = North:
  day  rate    SE  df lower.CL upper.CL
305 0.4144 0.0741 462    0.2917    0.5889
274 0.1049 0.0214 462    0.0702    0.1568
```

```
year = 2021, area = North:
  day  rate    SE  df lower.CL upper.CL
305 0.7353 0.1258 462    0.5253    1.0292
274 0.1861 0.0361 462    0.1271    0.2725
```

```
year = 2020, area = North Central:
  day  rate    SE  df lower.CL upper.CL
305 0.2110 0.0474 462    0.1358    0.3279
274 0.0534 0.0131 462    0.0330    0.0864
```

```
year = 2021, area = North Central:
  day  rate    SE  df lower.CL upper.CL
305 0.3743 0.0783 462    0.2481    0.5647
```

```
274 0.0948 0.0216 462 0.0605 0.1483
```

```
year = 2020, area = Central:
```

```
day rate SE df lower.CL upper.CL
305 0.1989 0.0818 462 0.0886 0.4463
274 0.0503 0.0204 462 0.0227 0.1116
```

```
year = 2021, area = Central:
```

```
day rate SE df lower.CL upper.CL
305 0.3529 0.1455 462 0.1569 0.7936
274 0.0893 0.0361 462 0.0403 0.1978
```

```
year = 2020, area = Southeast:
```

```
day rate SE df lower.CL upper.CL
305 0.3959 0.0725 462 0.2762 0.5675
274 0.1002 0.0211 462 0.0662 0.1517
```

```
year = 2021, area = Southeast:
```

```
day rate SE df lower.CL upper.CL
305 0.7025 0.1151 462 0.5091 0.9693
274 0.1778 0.0339 462 0.1222 0.2587
```

```
year = 2020, area = Island Park:
```

```
day rate SE df lower.CL upper.CL
305 0.4263 0.0858 462 0.2870 0.6331
274 0.1079 0.0232 462 0.0707 0.1647
```

```
year = 2021, area = Island Park:
```

```
day rate SE df lower.CL upper.CL
305 0.7563 0.1402 462 0.5254 1.0887
274 0.1915 0.0377 462 0.1300 0.2819
```

```
Confidence level used: 0.95
```

```
Intervals are back-transformed from the log scale
```

```
pairs(emmeans(m, ~ day | year * area, at = list(day = c(d11,d10)),
data = ticks, type = "response"), infer = TRUE)
```

```
year = 2020, area = North:
```

```
contrast ratio SE df lower.CL upper.CL null t.ratio p.value
day305 / day274 3.95 0.739 462 2.74 5.71 1 7.343 <.0001
```

```
year = 2021, area = North:
```

```
contrast ratio SE df lower.CL upper.CL null t.ratio p.value
day305 / day274 3.95 0.739 462 2.74 5.71 1 7.343 <.0001
```

```
year = 2020, area = North Central:
```

```
contrast ratio SE df lower.CL upper.CL null t.ratio p.value
day305 / day274 3.95 0.739 462 2.74 5.71 1 7.343 <.0001
```

```
year = 2021, area = North Central:
```

```
contrast ratio SE df lower.CL upper.CL null t.ratio p.value
day305 / day274 3.95 0.739 462 2.74 5.71 1 7.343 <.0001
```

```
year = 2020, area = Central:
```

```

contrast      ratio    SE  df lower.CL upper.CL null t.ratio p.value
day305 / day274 3.95 0.739 462    2.74    5.71    1   7.343 <.0001

year = 2021, area = Central:
contrast      ratio    SE  df lower.CL upper.CL null t.ratio p.value
day305 / day274 3.95 0.739 462    2.74    5.71    1   7.343 <.0001

year = 2020, area = Southeast:
contrast      ratio    SE  df lower.CL upper.CL null t.ratio p.value
day305 / day274 3.95 0.739 462    2.74    5.71    1   7.343 <.0001

year = 2021, area = Southeast:
contrast      ratio    SE  df lower.CL upper.CL null t.ratio p.value
day305 / day274 3.95 0.739 462    2.74    5.71    1   7.343 <.0001

year = 2020, area = Island Park:
contrast      ratio    SE  df lower.CL upper.CL null t.ratio p.value
day305 / day274 3.95 0.739 462    2.74    5.71    1   7.343 <.0001

year = 2021, area = Island Park:
contrast      ratio    SE  df lower.CL upper.CL null t.ratio p.value
day305 / day274 3.95 0.739 462    2.74    5.71    1   7.343 <.0001

```

Confidence level used: 0.95

Intervals are back-transformed from the log scale

Tests are performed on the log scale

What about the *difference* in the expected densities (i.e., marginal effects)?

```

tmp <- emmeans(m, ~ day | year * area, at = list(day = c(d11,d10)),
  data = ticks, type = "response")
pairs(regrid(tmp, type = "response"), infer = TRUE)

```

```

year = 2020, area = North:
contrast      estimate    SE  df lower.CL upper.CL t.ratio p.value
day305 - day274   0.309 0.0653 462   0.1812   0.438   4.739 <.0001

year = 2021, area = North:
contrast      estimate    SE  df lower.CL upper.CL t.ratio p.value
day305 - day274   0.549 0.1130 462   0.3271   0.771   4.860 <.0001

year = 2020, area = North Central:
contrast      estimate    SE  df lower.CL upper.CL t.ratio p.value
day305 - day274   0.158 0.0396 462   0.0799   0.235   3.985 0.0001

year = 2021, area = North Central:
contrast      estimate    SE  df lower.CL upper.CL t.ratio p.value
day305 - day274   0.280 0.0667 462   0.1485   0.411   4.193 <.0001

year = 2020, area = Central:
contrast      estimate    SE  df lower.CL upper.CL t.ratio p.value
day305 - day274   0.148 0.0642 462   0.0224   0.275   2.314 0.0211

year = 2021, area = Central:
contrast      estimate    SE  df lower.CL upper.CL t.ratio p.value

```

```
day305 - day274    0.264 0.1143 462    0.0389    0.488    2.305    0.0216
```

year = 2020, area = Southeast:

```
contrast      estimate      SE  df lower.CL upper.CL t.ratio p.value
day305 - day274    0.296 0.0632 462    0.1714    0.420    4.676 <.0001
```

year = 2021, area = Southeast:

```
contrast      estimate      SE  df lower.CL upper.CL t.ratio p.value
day305 - day274    0.525 0.1043 462    0.3197    0.730    5.032 <.0001
```

year = 2020, area = Island Park:

```
contrast      estimate      SE  df lower.CL upper.CL t.ratio p.value
day305 - day274    0.318 0.0743 462    0.1724    0.464    4.287 <.0001
```

year = 2021, area = Island Park:

```
contrast      estimate      SE  df lower.CL upper.CL t.ratio p.value
day305 - day274    0.565 0.1247 462    0.3198    0.810    4.529 <.0001
```

Confidence level used: 0.95

Here is the average marginal effect for each year (i.e., averaging across areas).

```
emmeans(regrid(pairs(regrid(tmp, type = "response"))), ~year)
```

```
year estimate      SE  df lower.CL upper.CL
2020    0.246 0.0467 462    0.154    0.338
2021    0.436 0.0780 462    0.283    0.590
```

Results are averaged over the levels of: area

Confidence level used: 0.95

Finally consider a comparison of areas.

```
pairs(emmeans(m, ~area | year, at = list(day = d10), type = "response",
  data = ticks), adjust = "none")
```

year = 2020:

```
contrast      ratio      SE  df null t.ratio p.value
North / North Central    1.964 0.453 462    1    2.930 0.0036
North / Central          2.084 0.854 462    1    1.792 0.0738
North / Southeast        1.047 0.201 462    1    0.238 0.8123
North / Island Park       0.972 0.196 462    1   -0.140 0.8891
North Central / Central   1.061 0.457 462    1    0.137 0.8908
North Central / Southeast 0.533 0.122 462    1   -2.758 0.0060
North Central / Island Park 0.495 0.118 462    1   -2.951 0.0033
Central / Southeast       0.502 0.207 462    1   -1.672 0.0952
Central / Island Park     0.467 0.193 462    1   -1.842 0.0661
Southeast / Island Park   0.929 0.186 462    1   -0.370 0.7119
```

year = 2021:

```
contrast      ratio      SE  df null t.ratio p.value
North / North Central    1.964 0.453 462    1    2.930 0.0036
North / Central          2.084 0.854 462    1    1.792 0.0738
North / Southeast        1.047 0.201 462    1    0.238 0.8123
North / Island Park       0.972 0.196 462    1   -0.140 0.8891
North Central / Central   1.061 0.457 462    1    0.137 0.8908
```

North Central / Southeast	0.533	0.122	462	1	-2.758	0.0060
North Central / Island Park	0.495	0.118	462	1	-2.951	0.0033
Central / Southeast	0.502	0.207	462	1	-1.672	0.0952
Central / Island Park	0.467	0.193	462	1	-1.842	0.0661
Southeast / Island Park	0.929	0.186	462	1	-0.370	0.7119

Tests are performed on the log scale

Due to an absence of interactions involving area, neither year or day matter.

```
pairs(emmeans(m, ~area, type = "response", data = ticks), adjust = "none")
```

contrast	ratio	SE	df	null	t.ratio	p.value
North / North Central	1.964	0.453	462	1	2.930	0.0036
North / Central	2.084	0.854	462	1	1.792	0.0738
North / Southeast	1.047	0.201	462	1	0.238	0.8123
North / Island Park	0.972	0.196	462	1	-0.140	0.8891
North Central / Central	1.061	0.457	462	1	0.137	0.8908
North Central / Southeast	0.533	0.122	462	1	-2.758	0.0060
North Central / Island Park	0.495	0.118	462	1	-2.951	0.0033
Central / Southeast	0.502	0.207	462	1	-1.672	0.0952
Central / Island Park	0.467	0.193	462	1	-1.842	0.0661
Southeast / Island Park	0.929	0.186	462	1	-0.370	0.7119

Results are averaged over the levels of: year

Tests are performed on the log scale

A Mixed Effects Generalized Additive Model

More recently we have been incorporating a random effect for GMU. A Poisson GLMM still shows over-dispersion, but unfortunately the **mgcv** package cannot handle a mixed effects negative binomial model. So instead we have been accounting for the over-dispersion by including a per-observation random effect which is effectively what the negative binomial distribution does.

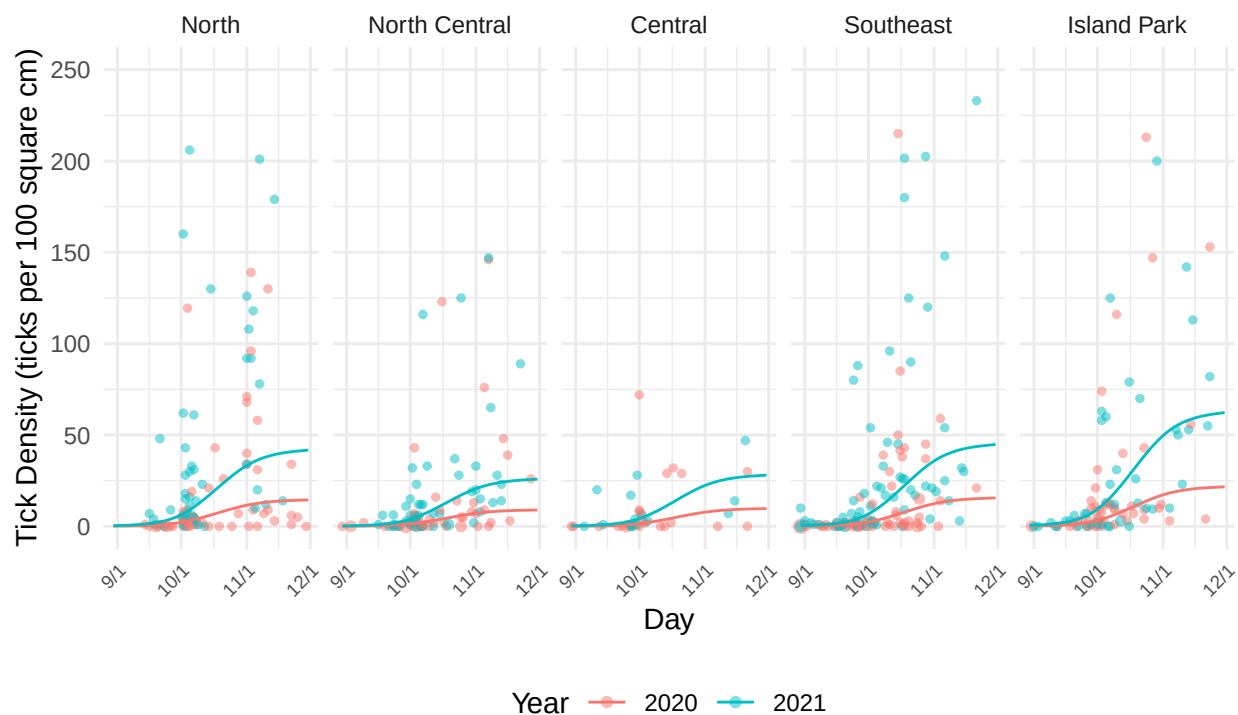
```
ticks <- ticks %>% mutate(obs = 1:n())
m <- gamm(ticks ~ offset(log(size)) + s(day) + year + area,
  family = poisson, random = list(gmu = ~1, obs = ~1), data = ticks)
```

Maximum number of PQL iterations: 20

Note the use of `m[["gam"]]` in predict below. This is a bit of a hack to get it to work with predict.

```
d$yhat <- predict(m[["gam"]], newdata = d, type = "response")
p <- ggplot(ticks, aes(x = day, y = ticks/size*100, color = year)) +
  geom_count(alpha = 0.5) + scale_size_area(max_size = 2) + theme_minimal() +
  facet_wrap(~ area, ncol = 5) +
  labs(x = "Day", y = "Tick Density (ticks per 100 square cm)", color = "Year") +
  guides(size = "none") +
  scale_x_continuous(breaks = c(d09, d10, d11, d12),
    labels = c("9/1", "10/1", "11/1", "12/1")) + ylim(c(0,250)) +
  theme(legend.position = "bottom",
    axis.text.x = element_text(angle = 45, size = 7, hjust = 1)) +
  geom_line(aes(y = yhat), data = d)
plot(p)
```

Warning: Removed 1 row containing non-finite outside the scale range (``stat_sum()``).



Inferences using this model can be done the same way. For example, we can look at the rate ratios for the effect of year.

```
emmmeans(m, ~year | area, at = list(day = d10),
  type = "response", offset = log(100), data = ticks)
```

area = North:

year	rate	SE	df	lower.CL	upper.CL
2020	2.20	0.686	462	1.190	4.06
2021	6.33	1.955	462	3.448	11.61

area = North Central:

year	rate	SE	df	lower.CL	upper.CL
2020	1.36	0.365	462	0.803	2.31
2021	3.92	1.018	462	2.353	6.53

area = Central:

year	rate	SE	df	lower.CL	upper.CL
2020	1.48	0.488	462	0.770	2.83
2021	4.25	1.421	462	2.203	8.20

area = Southeast:

year	rate	SE	df	lower.CL	upper.CL
2020	2.36	0.570	462	1.467	3.79
2021	6.79	1.601	462	4.276	10.79

area = Island Park:

year	rate	SE	df	lower.CL	upper.CL
2020	3.28	0.877	462	1.940	5.54
2021	9.45	2.525	462	5.587	15.97

Confidence level used: 0.95

Intervals are back-transformed from the log scale

```
pairs(emmeans(m, ~year | area, at = list(day = d10),
  type = "response", offset = log(100), data = ticks), reverse = TRUE, infer = TRUE)
```

area = North:

contrast	ratio	SE	df	lower.CL	upper.CL	null	t.ratio	p.value
year2021 / year2020	2.88	0.427	462	2.15	3.85	1	7.130	<.0001

area = North Central:

contrast	ratio	SE	df	lower.CL	upper.CL	null	t.ratio	p.value
year2021 / year2020	2.88	0.427	462	2.15	3.85	1	7.130	<.0001

area = Central:

contrast	ratio	SE	df	lower.CL	upper.CL	null	t.ratio	p.value
year2021 / year2020	2.88	0.427	462	2.15	3.85	1	7.130	<.0001

area = Southeast:

contrast	ratio	SE	df	lower.CL	upper.CL	null	t.ratio	p.value
year2021 / year2020	2.88	0.427	462	2.15	3.85	1	7.130	<.0001

area = Island Park:

contrast	ratio	SE	df	lower.CL	upper.CL	null	t.ratio	p.value
year2021 / year2020	2.88	0.427	462	2.15	3.85	1	7.130	<.0001

Confidence level used: 0.95

Intervals are back-transformed from the log scale

Tests are performed on the log scale